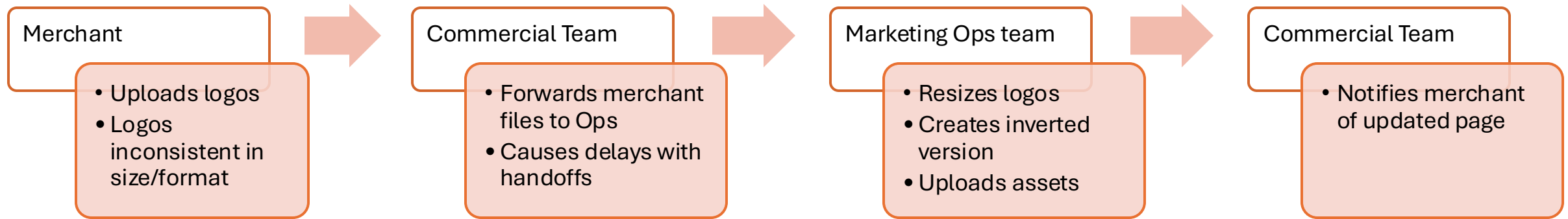




Tanya Irani —

Case Study

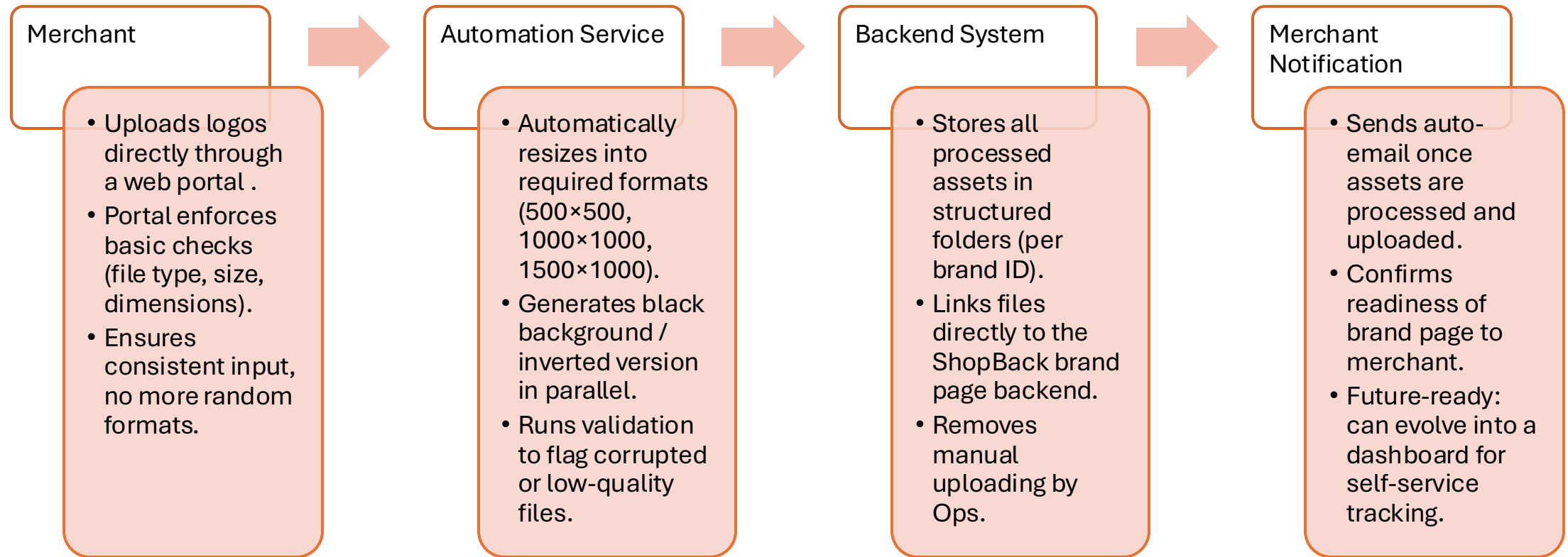
*Intern, Operations Automation,
ShopBack*



Challenges

- Manual, repetitive tasks → resizing + inversion
- Slow turnaround → delays brand activation
- Too many handoffs → inefficiency
- Errors → inconsistent quality
- Not scalable → can't handle growth

“Manual, repetitive workflows in logo onboarding cause delays, errors, and make the process unscalable.”



“Automation streamlines onboarding, reduces manual work, and enables scale.”

Key features

- Accepts only supported formats (PNG/JPG)
- Enforces dimension and file-size checks
- Provides clean, standardized inputs for downstream automation
- Auto-names files with **brand_id** for traceability
- Stores metadata (upload time, merchant ID) for Ops monitoring

SIMPLE WEB PORTAL

Pros: Quick to prototype and iterate
Easy for merchants to use — just upload and go
Allows custom validation rules (file type, size, dimensions)

Cons : Requires basic hosting and ongoing maintenance

Reasoning:
This option balances speed and flexibility. It can be developed as a lightweight MVP, tested quickly with Ops, and iterated on without heavy engineering investment.



FULL INTEGRATION WITH ShopBack CMS

Pros : Seamless in the long run → directly part of merchant onboarding
Eliminates the need for a separate tool

Cons: Heavy engineering lift → not suited for MVP

Assumption:
I am assuming ShopBack uses an internal backend or content management system (CMS) to manage merchant brand pages and assets. Integrating this workflow directly into that system would be the ideal end-state once the process is proven.

EXTERNAL FILE UPLOAD SERVICE

Pros: Very quick setup (days, not weeks)
Familiar to many users

Cons: Minimal control over validation
Files still inconsistent; Ops workload remains high

Reasoning:
Does not address the real issue of inconsistent inputs. It does not solve the core issue of **standardizing inputs**

LOW – CODE TOOL

Pros: Very fast deployment with drag-and-drop design
Easy to integrate with spreadsheets or databases

Cons: Limited flexibility for custom validation (e.g., precise image dimension checks)
Dependent on external vendor; harder to adapt for ShopBack’s needs

Reasoning:
Good for proof-of-concept or very small-scale pilots, but not ideal for ShopBack’s long-term needs where we need precise control over validation and scalability.

Recommendation: Start with a Simple Web Portal MVP for quick wins, and plan integration into ShopBack’s internal backend system once validated.

Key features

- Resizes logos automatically into required formats (500×500, 1000×1000, 1500×1000).
- Generates alternative logo on black background.
- Runs quality checks (rejects corrupted/too-small files).
- Outputs standardized files + metadata for downstream use.

LOW-CODE AUTOMATION TOOL

Pros : Fast deployment with drag-and-drop flows
Easy to connect with cloud storage or email

Cons: Weak at image-specific tasks (e.g., brand-safe inversion)
Doesn't scale well for large merchant volumes
Limited support for structured metadata generation

Reasoning:
Fine for proof-of-concept, but not sustainable for production. **Image quality and validation** risks remain.

OFF-THE-SHELF IMAGE PROCESSING LIBRARY / API

Pros: Very quick to implement, APIs already handle resizing and variants
Includes CDN options for delivery (bonus if needed later)

Cons: Vendor lock-in, recurring cost per file
Limited ability to enforce *ShopBack-specific validation rules*
Metadata integration may be partial / manual

Reasoning:
Good for MVP speed, but Ops would lose **fine-grained control**. Costs can grow with scale.

IN HOUSE SCRIPT

Pros: Full flexibility → can enforce brand rules (e.g., formats, dimensions)
Easy to extend (e.g., metadata manifest, new formats later)
Can be tightly integrated with portal + backend

Cons: Requires some initial coding + maintenance

Reasoning:
Best balance of flexibility and control. Ensures outputs are predictable and fully traceable. Fits ShopBack's need for **scalability + custom validation**.

Recommendation: Start with an **in-house automation script/service** (e.g., Python + Pillow or OpenCV).
Generate all file variants + metadata manifest automatically. Later, explore **API integration (Cloudinary/Imgix)** if delivery/CDN features become important.

Key features

- Stores all processed logo assets in a structured and traceable way (e.g., by brand ID).
- Links files directly to the ShopBack brand page backend.
- Ensures idempotency (no duplicates when merchants re-upload).
- Removes manual uploading by Ops.

DATABASE-FIRST STORAGE

- Pros:** Centralized — everything in one system (data + files)
Easier to query metadata directly
- Cons:** Inefficient for large image files → performance and cost issues
Harder to integrate with content delivery (CDN)
- Reasoning:**
Not recommended — storing large binary files directly in databases slows things down and adds complexity without clear benefits.

HYBRID APPROACH (STORAGE + METADATA DB)

- Pros:** Assets stored in S3/GCP with a lightweight database holding metadata (brand_id, size, upload time, checksum, etc.)
Provides full traceability and fast queries
Easy to extend into APIs later
- Cons:** Slightly more setup than simple storage-only approach
- Reasoning:**
This balances scalability and traceability. Ops can search and troubleshoot quickly using metadata, while engineers get reliable file storage.

INTEGRATION WITH EXISTING CMS

- Pros:** Seamless merchant brand page integration
Eliminates the need for a separate file-linking workflow
- Cons:** Requires heavy engineering effort upfront
May not allow for flexible naming conventions or metadata tracking
- Assumption:**
I am assuming ShopBack has an internal CMS that already manages merchant brand pages. Integrating directly here would be the long-term end state, but not ideal for MVP due to engineering overhead.

FILE SYSTEM / CLOUD STORAGE

- Pros:** Scalable, secure storage with folder structure per brand
Simple integration with existing backend
- Cons:** Requires some backend work for indexing and linking to brand records
Access controls must be configured carefully
- Reasoning:**
This is the industry-standard approach. Cloud storage provides scalability and reliability, while keeping files neatly organized by brand ID. Minimal Ops involvement after setup.

Recommendation :Start with **cloud storage + lightweight metadata database**. **Example:** Use AWS S3 with a Postgres table storing brand_id, file paths, checksums, and status. This removes Ops manual uploads immediately, ensures every file is traceable, and creates a clean integration point for the ShopBack brand page backend. In the long run, migrate to direct integration with ShopBack’s CMS once the process is proven.

Step 4: Merchant Notification

Key features

- Automatically informs merchants when their brand assets are processed and live.
- Reduces Ops effort of manual follow-ups.
- Provides merchants with traceability (which files were generated, where they are stored).
- Ensures consistent, professional communication.

EMAIL NOTIFICATIONS



- Pros:** Simple to implement (merchants already use email daily)
Can attach processed assets or links directly
Low barrier to adoption → no new tools for merchants
- Cons:** Limited tracking (can't easily see if merchant read/acknowledged)
- Reasoning:** Best option for MVP. Quick to set up, low learning curve for merchants, and builds trust with clear confirmation.

MERCHANT DASHBOARD / PORTAL INTEGRATION

- Pros:** Merchants can log in and check status anytime
Eliminates dependency on email
Allows richer features in future (e.g., re-upload, analytics)
- Cons:** Requires more development effort and UI maintenance
May not be necessary for early rollout
- Reasoning:** Ideal long-term solution for scalability and self-service. Too heavy for an MVP but should be considered in future roadmap.

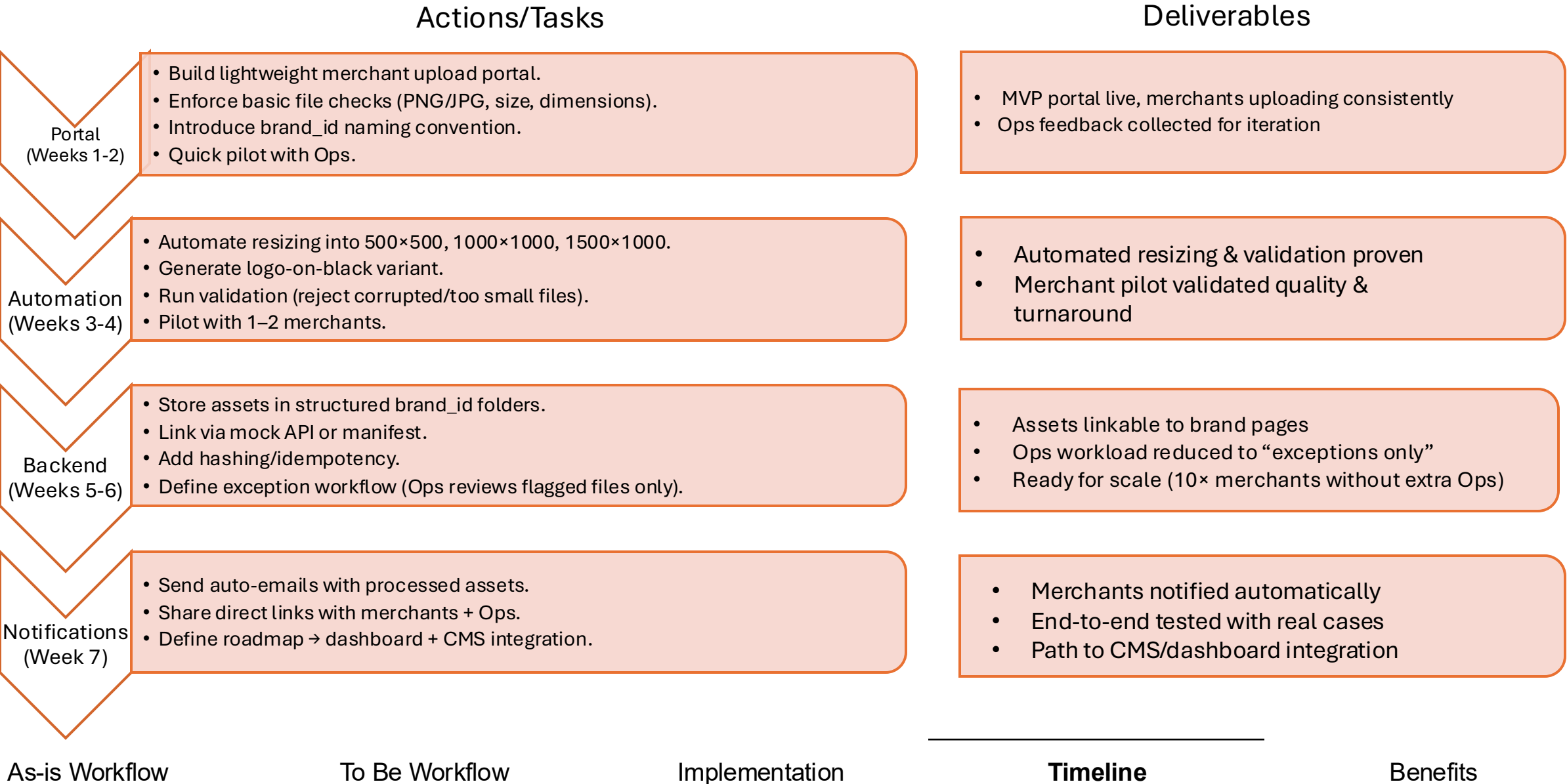
INTERNAL TEAM NOTIFICATIONS

- Pros:** Fast alerts to internal Ops/Commercial teams
Good for monitoring and troubleshooting
- Cons:** Not merchant-facing → merchants would still wait for manual updates
- Reasoning:** Useful as a supplement for Ops visibility, but does not replace merchant notifications.

SMS / WHATSAPP ALERTS

- Pros:** Immediate delivery, merchants unlikely to miss it
- Cons:** Costly at scale
Not ideal for sending multiple files or detailed info
- Reasoning:** Works well for urgent alerts, but not practical for regular asset delivery.

Recommendation: Start with Email Notifications for MVP — reliable, simple, and merchant-friendly.
Later, evolve into a merchant-facing dashboard where they can view statuses and download assets directly.



Benefits	Future Improvements	AI Enhancements
• Faster Turnaround: Hours/days → minutes. • Consistency & Quality: Standardized outputs, fewer errors. • Reduced Ops Workload: Ops focus only on flagged exceptions. • Scalability: System can handle growing merchant volumes without extra headcount. • Better Traceability: Brand ID, file naming, and metadata ensure full audit trail. • Improved Merchant Experience: Merchants notified instantly, brand pages go live faster.	• CMS Integration: Full embedding into ShopBack’s backend so the process is part of merchant onboarding. • Merchant Dashboard: Move from email → self-service dashboard for real-time status and asset downloads. • Ops Analytics: Use collected metadata (upload time, file failures, exceptions) to spot bottlenecks and train Ops.	• AI-driven Quality Checks: Detect low-quality, blurry, or off-brand logos automatically, not just based on size. • Smart Background Handling: AI vision models can distinguish between logo vs. background and auto-generate clean, brand-safe variants. • Auto-tagging & Classification: Use AI to tag assets (e.g., “logo horizontal,” “logo with tagline”) for better searchability in backend. • Predictive Ops Support: AI can flag merchants that frequently upload non-compliant assets, so Ops can pre-emptively engage. • AI Chatbot for Merchants: Allow merchants to query “Where’s my logo?” or “What failed?” directly, reducing Ops tickets.